



Graphic Era
HILL UNIVERSITY

Established by an Act of the State Legislature of Uttarakhand (Admission Session 12 of 2011)
University under section 2(f) of UGC Act, 1956

GEHU/02E/9.1.3

End Term (Even) Semester Regular/Back Examination May 2026

Roll no. 2321109

Name of the Program and semester: BCA/BCA(AI/DS)/B.Sc. IT 6th

Name of the Course: Computer Graphics

Course Code: TBC 604(1)/TBD 604(a)/ TBI 602

Time: 3-hour

Maximum Marks: 100

Note:

- (i) All the questions are compulsory.
- (ii) Answer any two sub questions from a, b and c in each main question.
- (iii) Total marks for each question are 20 (twenty).
- (iv) Each sub-question carries 10 marks.

- Q1. (2X10=20 Marks)
- a. Analysis the performance of 2D successive scaling and transformation. (CO3)
 - b. How 3D rotation is performed about x-axis, y-axis and z-axis along with x' , y' and z' values. (CO4)
 - c. What is GKS (Graphical Kernel System) in computer graphics? Explain its features and how it helps in creating graphical applications. (CO6)

- Q2. (2X10=20 Marks)
- a. How hidden surface detection performed? Explain the single and multiple surfaces methods to identify the hidden surfaces. (CO4)
 - b. Line AB coordinates are (10, 10) to (18, 19), implements Bresenham line algorithm. (CO2)
 - c. Implement Mid-Point circle generation algorithm for radius 9. (CO2)

- Q3. (2X10=20 Marks)
- a. What is direct view storage tube display device? Explain also the shadow mask display device. (CO1)
 - b. Write steps for Boundary-Fill and Flood-Fill method for 4-connected pixels. (CO5)
 - c. Draw taxonomy of projection and explain with their characteristics and types. (CO6)

- Q4. (2X10=20 Marks)
- a. Write steps for Sutherland Hodgman polygon clipping algorithm. Now explain what are the entries come under the vertex output list. (CO4)
 - b. When the angles are perpendicular each other, same direction and opposite direction? Also find out the angle between A and B vectors. (CO2)
 - c. Explain the types of animation systems? How animation system is beneficial for computer graphics. (CO6)

- Q5. (2X10=20 Marks)
- a. Write steps for Liang Barsky line clipping algorithm? How to compute the intersection point between boundary and given line? (CO4)
 - b. Perform 45-degree rotation on triangle (5, 5), (10, 5) and (5, 10) while keeping (4, 4) point fixed. (CO3)
 - c. Write steps for Digital differential analyzer line algorithm. (CO2)